CONSOLIDATED PROBATIONARY PERIOD REPORT

Development of a Machine Learning-Based Exploit Generation Module for Various Android OS File Systems

Consolidated report - Probationary period 1

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Repository: https://github.com/Nikeka-git/android-fs-ml-research

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Source commit: a50c495b30366b298fa8c7990772096c512773f9

Reviewed research/evidence baseline shown above. The refreshed submission tag also contains this report and its editable sources; no new runtime or ML result is claimed.

24 October 2026

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1 Project proposal

Official title: Development of a Machine Learning-Based Exploit Generation Module for Various Android OS File Systems.

The team's clarified research goal is to investigate ML/AI-assisted selection of existing research materials using Android device characteristics. The official title is retained, while the present probationary stage is a diagnostic and evidence-preparation stage. A trained selection model or a new exploit-generation system is not delivered in this version.

Problem and research question

Research artifacts can describe different device contexts, revisions, operating assumptions and evidence levels. A source file, a diagnostic text and a reproduced observation are distinct kinds of material. Before evaluating a future intelligent component, the team needs an inspectable prototype and a traceable record of what each artifact establishes.

The current-stage question is: can the available Android diagnostic prototype, research snapshots and source records be organized so that an independent reader can identify their origin, scope, limitations and individual contributors? Future effectiveness claims require a separate agreed experimental study.

Objectives and deliverables

Objective	Deliverable in this version	State
Organize the technical baseline	Two registered modules, module descriptions and source manifests	Prepared
Preserve research materials	204-file import index and a registry of 12 public diagnostic texts	Prepared
Define research documentation	Data-preparation rules, evaluation methodology and limitations	Prepared
Trace team work	Commit/PR links, 23 real task records and dated screenshots	Prepared
Demonstrate future ML effectiveness	Trained model, independent evaluation and device-level results	Not yet evidenced

Scope and completion boundary

This report consolidates the current artifacts and plans. It records preserved tests as historical results and recent verification as integrity checking. Android builds, device trials, PoC execution and ML training were not performed while preparing the submission. The three PoC source groups remain one research module, in the repository structure selected by the team.

The project repository and submission version are accessible at the public repository and prob-period-1. The technical and evidence baseline reviewed for this report is a50c495. The final tag also includes this report and its editable sources in a documentation-only revision.

2 Evidence of relevance

Android security research spans application behavior, operating-system services, kernel interfaces and storage. These layers should not be conflated. AOSP documents Binder as an IPC system and SELinux as a mandatory-access-control mechanism; a denied diagnostic operation or a Binder observation cannot by itself establish a file-system defect. [2] [1]

The ext4 documentation and the original F2FS paper describe different file-system designs and terminology. They provide background for selecting and describing future research contexts, rather than evidence that the team has tested either file system on a particular Android device. [4] [3]

Why the present preparation matters

- Traceability makes it possible to distinguish an imported artifact from a result produced by the current team and revision.
- Dataset documentation exposes source, annotation and usage limitations before model conclusions are drawn.
- A visible separation between a local diagnostic prototype and a future research goal makes the current stage reviewable.
- Linked tasks, commits and evidence let the supervisor and committee inspect individual contributions.

These priorities are consistent with the emphasis on data documentation in Datasheets for Datasets and on integrity and verification in the NIST Secure Software Development Framework. They are an academic rationale for this stage, not a claim of full conformance or certification. [6] [5]

Project-specific evidence and practical value

Available material	Value to this stage	Limit of the evidence
Android/Kotlin/JNI/C++ snapshot	Concrete diagnostic prototype source to inspect	No fresh post-import build or device run
Three groups in one PoC research module	Preserved, navigable source and report collection	Presence does not prove original authorship or execution
12-source registry, 46,570 bytes	Pinned origins, sizes and SHA-256 values	Small diagnostic-text corpus; labels await independent review
Board and contributor history	Inspectable work allocation and delivered artifacts	Integration is not independent peer review

The immediate beneficiaries are the research team and academic reviewers. The current contribution is an auditable foundation for later analysis. No market-size estimate, deployment adoption, accuracy improvement or device coverage is inferred from the available materials.

3 Literature review

Android platform context

AOSP's Binder overview explains the IPC boundary used by Android services. It supplies the correct context for the diagnostic baseline. AOSP's SELinux documentation explains why permissions and enforced access-control policy belong in the interpretation of an observation. Together these sources support explicit context recording rather than assumptions based only on a function name or a nominal platform version. [2] [1]

File-system background

Lee, Sim, Hwang and Cho introduce F2FS as a file system designed around flash-storage characteristics. The paper offers design background and reports experiments on the systems evaluated by its authors. The Linux kernel's ext4 documentation provides a maintained reference for ext4 data structures and algorithms. Neither source demonstrates the configuration, security state or behavior of a team device. [3] [4]

Data and model reporting

Gebru and colleagues propose documenting dataset motivation, composition, collection, intended uses and limitations. This is directly relevant to the 12-source registry: source hashes identify bytes, while labels and representativeness require separate justification. Mitchell and colleagues propose model cards to communicate intended use, evaluation and limitations. The latter is a future reporting framework here, because the tagged repository does not provide a trained Android selection model. [6] [7]

Development process and related infrastructure

NIST SSDF provides high-level practices for integrating secure development into a software lifecycle. This report uses traceability and preserved evidence as relevant process principles; it does not map every SSDF practice or assert compliance. The NVD overview describes a vulnerability-data resource. The syzkaller repository describes a kernel-testing project, and its report tests provide context for the public diagnostic fixtures used by the team. These resources have different purposes and are not interchangeable benchmarks. [5] [8] [9] [10]

Synthesis and research gap

The cited work motivates context-aware interpretation, documented data and explicit evaluation boundaries. The present gap is project-specific: the repository still needs independently checked annotations, available prototype outputs, an agreed test environment and a complete evaluation record. The literature review therefore supports preparation and cautious interpretation; it supplies no evidence that the team's intended ML goal has already been achieved.

The review uses selected primary sources rather than claiming to be a systematic review. Search coverage, publication completeness and superiority over every related system are not established. Bibliographic links are provided in References.

4 Existing system comparison

The comparison concerns documented purpose, inputs and evidence, rather than performance. No common benchmark was executed. NVD and syzkaller provide useful reference points, but neither is presented as a product with the same scope as this student project.

System / resource	Documented role	Relation to this project	Comparison limit
NVD	Vulnerability metadata resource [8]	Reference context for structured public records	Catalogue entries do not prove a team device observation
syzkaller project	Kernel-testing infrastructure [9]	Upstream context for public diagnostic materials	Its functionality and results are not the team prototype's results
syzkaller report fixtures	Inputs exercised by upstream report tests [10]	12 pinned public texts are indexed in the team registry	Fixtures are not newly collected Android-device experiments
Local binder-diagnostics	Android source snapshot for local simulation and Binder diagnostics	Current inspectable prototype	Post-import build/runtime and multi-filesystem behavior unverified
Local android-poc-research	One preserved source/report research module	Current research artifact collection	No claim that every imported source builds or runs

Resulting positioning

The project's immediate role is to connect a concrete diagnostic source baseline with a transparent research record. It can refer to upstream materials without claiming their original authorship. Its future intelligent component remains a research objective. A credible later comparison must first define a shared task, observation unit, environment and evaluation material; this report does not invent such results.

Repository evidence: prototype description, research-module description and diagnostic-text registry.

5 Technology comparison

The technologies below are compared against the tasks already visible in the repository. This is a record of the current stack and its tradeoffs, rather than evidence that a new technology was implemented or benchmarked during report preparation.

Technology	Role in the current version	Reason / tradeoff	Verification state
Kotlin + Android resources	Application UI and diagnostic coordination	Platform-oriented UI; requires an Android environment	Source inspected
JNI + C++17	Native boundary and diagnostic/simulation implementation	Connects Kotlin to native code; increases cross-language review needs	Source snapshot; no new build
Gradle / CMake	Android/native build configuration	Versioned configuration; local SDK/NDK still required	Configuration observed
Python standard library	Manifest/schema and source-integrity checking	Readable offline checking without executing input texts	Local integrity check passed
JSON + CSV manifests	Source records, indexes and task snapshots	Machine-readable and reviewable; correctness still needs review	Published and parsed
Git + GitHub Issues/Project	Version, contribution and task records	Links artifacts to changes; Project visibility is separate	Public repository; private Project
Editable SVG / Mermaid	Architecture and interaction documentation	Text/vector sources support revision and review	Sources included with this report

Configuration and alternatives

The Android module records Java 17, compileSdk 36, NDK 28.2.13676358 and CMake 3.22.1. These are declared configuration values, not a list of validated devices. Local SDK paths and generated binaries were excluded from the imported snapshot. The source description and configuration files remain the authoritative record.

A single-language prototype could reduce interface complexity but would not describe the existing JNI-based baseline accurately. A database could support richer querying later, while the present JSON/CSV manifests keep this small evidence set reviewable. Raster images are useful preserved artifacts; editable SVG and Mermaid are better suited to future diagram revision. These are qualitative tradeoffs with no measured cost or speed claim.

No new ML library or candidate algorithm is selected in this report. Model choice remains open until the research question, data and evaluation protocol are agreed. File-system background is recorded through the original F2FS study and official ext4 documentation without asserting deployment support. [3] [4]

6 Requirements specification

The requirements distinguish existing artifacts, documented plans and unverified behavior. Acceptance is based on inspectable evidence. The list is a current-stage research specification; it does not replace a full official university marking rubric.

ID	Requirement and acceptance evidence	Current status
FR-01	Provide prototype source and its module description at the tagged URL	Source available
FR-02	Represent local simulation and diagnostic behavior with distinct status/results in the UI source	Observed in source; runtime unverified
FR-03	Register both research modules with explicit scope and limitations	Documented
FR-04	Preserve the three PoC source groups in one module and publish the 204-file integrity index	Prepared; import verification recorded
FR-05	Record pinned origins, size and SHA-256 for 12 diagnostic texts	Prepared
FR-06	Link individual completed work to real commits/PRs and task IDs	Prepared
FR-07	Deliver one report with 11 ordered sections, contents, editable diagrams and accessible links	Prepared by this documentation revision
FR-08	Provide a working trained Android selection component with independently measured quality	Future work; not evidenced

Non-functional and evidence requirements

ID	Requirement	Acceptance boundary
NFR-01	Traceability	Record source revision, artifact path, contributor and verification type
NFR-02	Integrity	Retain original source bytes and compare recorded SHA-256 values
NFR-03	Privacy and licensing	Keep applicable attribution; exclude private device dumps and personal raw logs from the public package
NFR-04	Reproducible reporting	Publish editable report/diagram sources and distinguish preserved reports from new runs
NFR-05	Accessible review	Public tagged links and dated task screenshots; separate Project access state
NFR-06	Honest uncertainty	Unavailable values remain unknown; absent runtime or model evidence is not reported as success

AOSP's SELinux description supports the need to record security context when interpreting unavailable observations. Dataset documentation and SSDF support the provenance and integrity requirements. These references explain the principles; satisfying a listed documentation requirement is not a certification of the entire system. [1] [6] [5]

Open decisions include the detailed target population, actual environment, independent annotation review, complete model-selection policy and the final evaluation material. These are recorded as pending rather than backfilled as completed requirements.

7 Architecture and diagrams

Current artifact architecture

The repository contains two independently described source modules, common documentation, research data registries, evidence and planning records. The shared directory is reserved for components with actual reuse; the presence of a directory does not establish a deployed service or integrated ML pipeline.

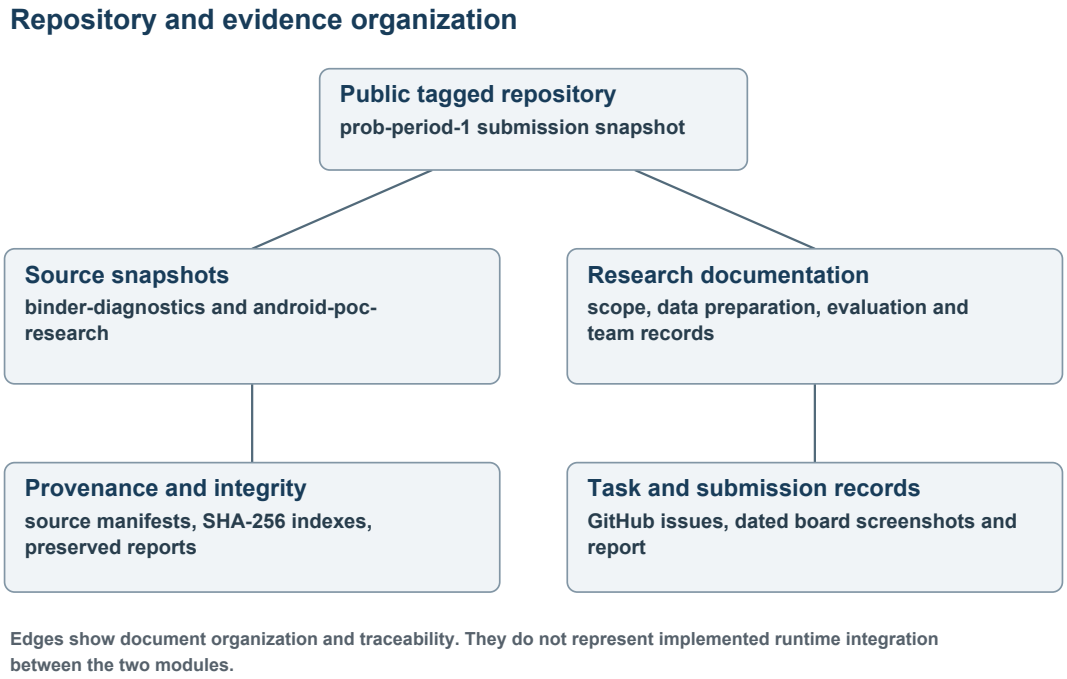


Figure 1. Editable repository/evidence organization. Edges express traceability, not runtime integration. Open supporting material

Editable diagram: SVG source. The earlier repository architecture is also editable as Mermaid in architecture.md.

Prototype boundary

The Android module contains a UI/coordinator layer, a Kotlin/native interface and native diagnostic/simulation code. Existing JVM, Android and native test sources are stored alongside it. The research snapshot module is a separate collection of source versions and reports. No implemented connection that turns those collections into a trained selection service is evidenced.

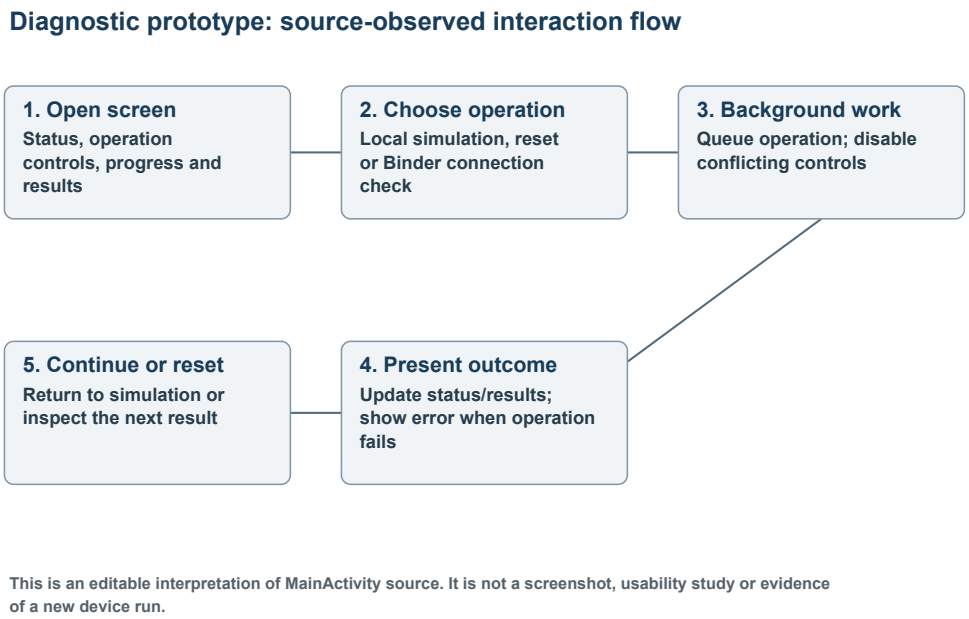


Figure 2. Source-observed diagnostic interaction. This diagram was prepared for the report; it is not a runtime capture. Open supporting material

Editable interaction diagram: SVG source. The supplied Assignment 3 system-flow PNG remains an original provided figure; its original editable diagram source was not supplied. The new SVGs document the current repository and do not claim to reconstruct that missing original.

Binder terminology follows the AOSP IPC description. Its role should remain distinct from the later file-system research question. [2]

8 UX and working prototype

The available prototype is the Android source module binder-diagnostics. Its README describes local simulation and Binder protocol diagnostics. A newly verified installable APK, hosted demonstration and current runtime screenshot are not available in the reviewed package. The link therefore provides an inspectable prototype source, with execution status stated explicitly.

Source-observed interaction

UI element / action	Behavior represented by the source	Evidence limit
Status and result fields	Display current operation, progress and textual results	Not a new observed device session
Discovery / full simulation	Initiate the existing local simulation path	Not an ML-selection result
Reset / return to simulation	Reset or switch the diagnostic session back to simulation	Runtime behavior still needs confirmation
Binder connection check	Initiate the existing device-diagnostic action	IPC observation, not file-system verification
Busy state / error handling	Queue work, disable conflicting controls and display an error when appropriate	Source inspection; no usability or robustness trial

The relevant source entry points are MainActivity.kt and activity_main.xml. Figure 2 records this interaction for review without presenting a fabricated application screenshot.

Prototype review criteria

- A reviewer can locate the application source, configuration, module description and test source directories from the tagged module.
- The module description identifies the current simulation/diagnostic scope and build-verification limitation.
- A later demonstration should record its actual code revision, environment, visible outputs and failures. No success screen is inferred from source alone.

Remaining UX evidence

There is no documented user study, accessibility audit, observed task-completion time or fresh installation test in this version. A later prototype review can check whether operation names, busy states, results and error explanations are understood by intended academic reviewers. Those checks remain planned. The source-level interaction diagram and linked module supply the currently accessible prototype material.

9 Verification and evaluation plan

Evidence actually available

Check / record	Observed result	Interpretation
Preserved JUnit XML	24 tests, 0 failures, 0 errors; report dated 15 Oct 2026	Historical simulation-domain result; not rerun for this report
Preserved lint report	0 errors, 1 warning	Historical static report; not a new clean build
PoC text import index	204 source texts with paths, sizes and SHA-256	Preservation evidence; not proof of execution
Diagnostic registry	12 sources, 46,570 bytes; 5 ext4 / 2 F2FS / 5 none reference labels	Labels imported from Assignment 3; independent review pending
Offline provenance verifier	12 records and 2 local artifacts checked; 0 exact duplicates	Schema/local hash check; no input execution or model test
Board screenshot PDF	13 pages inspected for coverage and readability	Evidence of captured task metadata, not technical effectiveness

Preserved result files are linked through the stored-evidence manifest. Recent checking used the repository provenance verifier in its local mode. That mode checks records and local artifact integrity; it does not fetch or execute diagnostic inputs.

Reported results that remain unreplicated

The jointly authored Assignment 3 contains parser and text-prototype performance claims. Its corresponding implementation, raw outputs, model and complete split ledger are not in this repository. Those claims remain documentary statements of the supplied report and are not promoted to independently verified project results. This consolidated report does not reconstruct missing trials or confidence intervals.

Next-stage validation gates

Gate	Planned evidence	Decision rule
V1: annotations and data	Independent label review, preserved disagreements and a versioned source ledger	Unresolved labels and unavailable sources remain visible
V2: reproducible text prototype	Implementation revision, inputs, saved outputs and actual split manifest	Separate development checks from independent evaluation
V3: Android prototype	Recorded environment and fresh build/runtime outcomes	Report failures and unavailable observations alongside successes
V4: future model reporting	Intended use, model identity, evaluation population and limitations	Do not claim quality without measured evidence
V5: general report review	Supervisor discussion record and requirement-to-evidence map	Resolve missing decisions before asserting final approval

The existing evaluation methodology separates source integrity, diagnostic classification, record completeness and future system evaluation. Missing data must not silently become a negative outcome; small data sets require absolute counts and clear denominators. Datasheets and model cards support this separation of provenance, intended use and measured performance. [6] [7]

This report adds documentation and diagrams only. No new Android/ACK experiment, PoC execution, model training, measured device coverage or effectiveness improvement is claimed.

10 Project management and risks

Task board and contribution evidence

The reconciled board contains **23 tasks: 12 Done, 8 In Progress and 3 Todo**. Completed records are linked to actual artifacts. Partial work remains open. Retrospective tasks register existing deliverables and do not manufacture historical development dates. The 24 October deadline on submission artifacts is a milestone date assigned during registration; unknown future deadlines remain unset.

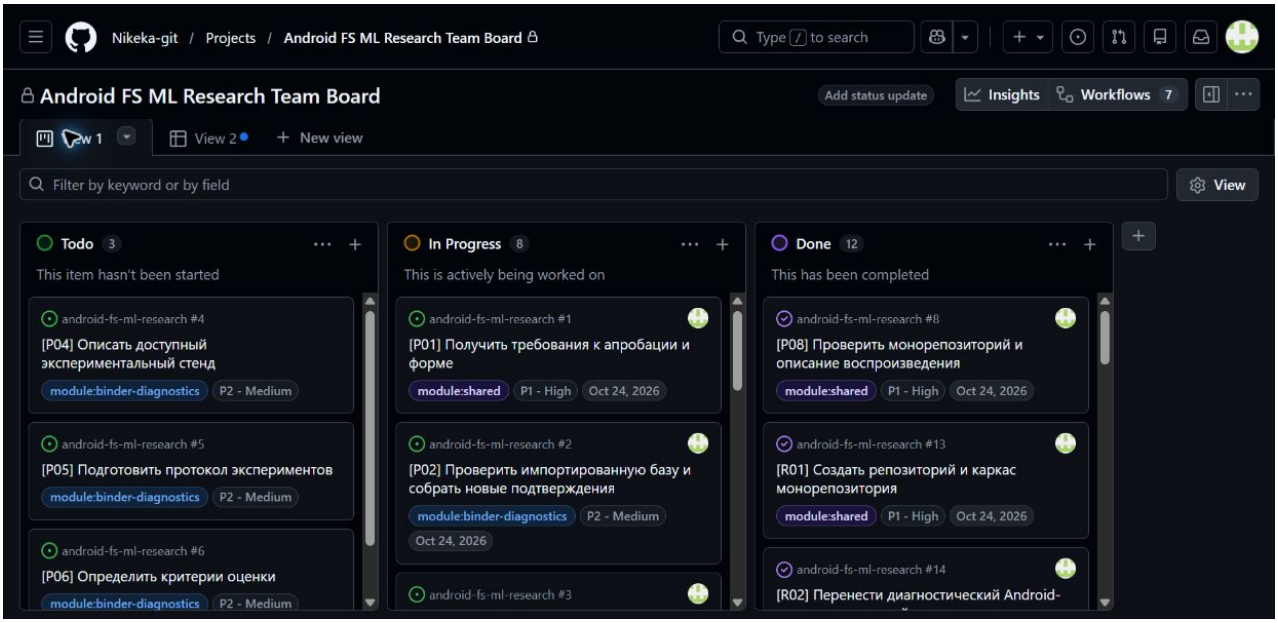


Figure 3. Actual GitHub board capture, 24 October 2026. Full-size task evidence is linked below. Open supporting material

The Project itself is private. The public issues, 23-task table and full-size board/task PDF remain publicly accessible. The latter shows assignees, priorities, deadlines, statuses and completed examples with acceptance criteria and commit/PR links.

Member and responsibility	Completed deliverables	Direct evidence
Tsybus Nikita Valentinovich; technical lead	Diagnostic baseline, monorepo organization, research import and submission documentation	#8, #13-#20; b23517a, 9b84849
Kassymgali Bakytzhan Zhenisuly; data/methodology and presentation	Source registry, provenance verification, data/evaluation documents and repository navigation	#23, #24; PR #21, PR #22
Galidullaeva Aliya Khantanirovna; analytical documentation	Two-page analytical summary of goal, modules, current evidence and limitations	#25; eb6dc9e

No YouTrack system was used. These are GitHub issue identifiers. The joint report retains shared authorship; an import commit does not establish original authorship of every pre-existing file. PR integration is evidenced, while independent reviews and review-driven revisions are not inferred.

Risk register and next actions

Risk	Consequence	Mitigation / next action	Owner
Scope drift between official title and working goal	Unclear acceptance standard	Record clarified scope with the supervisor	Nikita
Unverified runtime and missing prototype outputs	Demonstration or results cannot be reproduced	Obtain environment and actual run records	Nikita
Small corpus and unreviewed labels	Unsupported generalization	Review labels and record lineage/split before evaluation	Bakytzhan
Incomplete report review or verbal-only feedback	Academic decisions lack a written record	Confirm discussion points and revise documentation	Nikita with team
Private Project or changing tag	Committee sees inconsistent evidence	Keep public snapshots and use one frozen submission tag	Nikita
Sensitive raw logs / source attribution	Inappropriate public disclosure or attribution	Retain exclusions, origin records and license notices	All members

The next milestone is an agreed, inspectable research and evaluation package with fresh evidence where needed. The owners above describe next-stage responsibility; they are not claims that those future actions have already been completed. Detailed future dates and effort estimates are not fabricated. [5]

11 Supervisor feedback

Recorded status

Supervisor: **Zhaksylyk Kozhakhmet**. On 24 October 2026, the team representative reported that the supervisor had orally given agreement and confirmation in discussion. The representative did not supply exact wording, a meeting date or a written message.

This is a record of the representative's account of verbal feedback. The scope of each confirmation is not reconstructed. It is not presented as a quotation, signed approval, independent review of the experimental results or formal acceptance of every item in this report.

Feedback item	Available record	Status / follow-up
General verbal agreement / confirmation	Representative's statement supplied during report preparation	Recorded as reported verbal feedback
Exact comments and meeting date	Not supplied	Request a brief dated confirmation when available
Detailed scope and evaluation decisions	No decision-by-decision record supplied	Confirm the working goal, environment and evidence expectations
Revisions resulting from specific feedback	No precise comment-to-change mapping available	Do not assign existing commits to invented review requests

Agenda for the next documented consultation

- Confirm how the official generation title relates to the clarified selection-oriented research goal.
- Agree the current prototype demonstration and the evidence needed for a fresh build/runtime claim.
- Review the data/methodology package, remaining annotation questions and evaluation boundaries.
- Record specific comments, team actions and resulting revisions with dates and links.

The team can update this section when a dated feedback record is available. The presence of this section fulfils the supplied outline's ordering, while the completeness of supervisor evidence remains visible rather than replaced with invented quotations.

References

[1] Android Open Source Project. Security-Enhanced Linux in Android. Updated 2026-09-30. Accessed 24 October 2026. https://source.android.com/docs/security/features/selinux

[2] Android Open Source Project. Binder overview. Updated 2026-10-09. Accessed 24 October 2026. https://source.android.com/docs/core/architecture/ipc/binder-overview

[3] Changman Lee; Dongho Sim; Joo-Young Hwang; Sangyeun Cho. F2FS: A New File System for Flash Storage. 2015-02. 13th USENIX Conference on File and Storage Technologies (FAST 15), pp. 273-286. Accessed 24 October 2026. https://www.usenix.org/conference/fast15/technical-sessions/presentation/lee

[4] The Linux kernel development community. ext4 Data Structures and Algorithms. n.d. Accessed 24 October 2026. https://www.kernel.org/doc/html/latest/filesystems/ext4/

[5] Murugiah Souppaya; Karen Scarfone; Donna Dodson. Secure Software Development Framework (SSDF) Version 1.1: Recommendations for Mitigating the Risk of Software Vulnerabilities. 2022-02. NIST Special Publication 800-218. Accessed 24 October 2026. https://csrc.nist.gov/pubs/sp/800/218/final

[6] Timnit Gebru; Jamie Morgenstern; Briana Vecchione; Jennifer Wortman Vaughan; Hanna Wallach; Hal Daumé III; Kate Crawford. Datasheets for Datasets. 2021-12. Communications of the ACM, 64(12), pp. 86-92. Accessed 24 October 2026. https://arxiv.org/abs/1803.09010v8

[7] Margaret Mitchell; Simone Wu; Andrew Zaldivar; Parker Barnes; Lucy Vasserman; Ben Hutchinson; Elena Spitzer; Inioluwa Deborah Raji; Timnit Gebru. Model Cards for Model Reporting. 2019-01. Conference on Fairness, Accountability, and Transparency (FAT* '19). Accessed 24 October 2026. https://arxiv.org/abs/1810.03993v2

[8] Harold Booth; Doug Rike; Gregory A. Witte. The National Vulnerability Database (NVD): Overview. 2013-12-18. NIST ITL Bulletin. Accessed 24 October 2026. https://www.nist.gov/publications/national-vulnerability-database-nvd-overview

[9] The syzkaller project authors. syzkaller - kernel fuzzer (README). n.d. Accessed 24 October 2026. https://github.com/google/syzkaller/blob/master/README.md

[10] The syzkaller project authors. syzkaller report parser tests and test-data loading. n.d. Accessed 24 October 2026. https://github.com/google/syzkaller/blob/master/pkg/report/report_test.go

Appendix A. Evidence map for the committee

All repository links below use the submission tag. The reviewed research/evidence baseline is a50c495. The tag is refreshed once to include this consolidated report, its editable sources and navigation updates. Report preparation does not alter the imported module sources.

Report area	Artifact / direct tagged link	What it establishes
1-2: scope and relevance	Project scope; README	Official title, clarified goal and current state
3-5: research and comparison	Data preparation; Reference source register	Provenance of the documented materials and primary references
6: requirements and status	Project metadata; Task specification	Prepared artifacts, pending decisions and explicit acceptance criteria
7: editable architecture	Repository SVG; Interaction SVG	Editable report diagrams; documentation rather than runtime proof
8: prototype	Prototype source; Prototype README	Accessible source and build-verification boundary
9: preserved tests	JUnit XML; Lint text	Saved historical reports, not new runs
9: source integrity	204-file index; 12-source registry	Source identities, sizes and recorded hashes
10: board and team	Board/task PDF; Team record	Captured task metadata and commit/PR-backed contributions
11: supervisor feedback	Section 11 of this report	Reported verbal agreement, with missing written details identified
Editable report package	Report content JSON; PDF builder	Reviewable text, diagram references and reproducible report construction

Repository version and accessible materials

Repository: https://github.com/Nikeka-git/android-fs-ml-research.
Submission version: https://github.com/Nikeka-git/android-fs-ml-research/tree/prob-period-1.
Original reviewed baseline: a50c495.

Supporting source snapshots, the joint Assignment 3 and the analytical summary retain their original scope. Historical reports dated 15 October are preserved for traceability. They should not be mistaken for fresh measurements or a replacement for the consolidated current report.

Submission checklist

- Eleven sections follow the order supplied in the example, with a table of contents and References.
- The filename uses group SE-2421 and representative surname Tsybus.
- Prototype source, editable SVG/Mermaid diagrams and supporting evidence are linked.
- Verbal supervisor feedback is attributed to the representative; unavailable details remain explicit.
- The report and editable sources are published together in the refreshed prob-period-1 version.